Karthik Bhattaram

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Education

University of California, Santa Barbara — Santa Barbara
Master of Science in Computer Science — *March 2026 to March 2027*

- Graduate Courses: Computer Architecture, Computer Security, Foundation Models
- Researching machine learning and cybersecurity under Christopher Kruegel and Giovanni Vigna

University of California, Santa Barbara — Santa Barbara
Bachelor of Science in Computer Science — *September 2022 to March 2026*

- Selected Courses: Computer Architecture, Operating Systems, Computer Communication Networks
- Honors: Regents Scholar, College of Engineering Honors, Distinction in Major

Experience

Machine Learning Research Assistant — June 2026 – September 2026
Lawrence Berkeley National Lab — *Berkeley, California*

- Developed evaluations to compare network foundation models with traditional deep learning architectures
- Distilled larger models into decision trees that achieved similar accuracy with a greatly reduced footprint

Security Research Assistant — October 2025 – June 2026
Shellphish — *Santa Barbara, California*

- Built a benchmark of 8 Intrusion Detection Systems against 9,700+ C2 beacon samples spanning 10 frameworks
- Showed learned behavioral detectors achieve near perfect incident recall while signature-based methods fall short

Software Engineer Intern — June 2025 – September 2025
Arista Networks — *Santa Clara, California*

- Built C++ utilities to decode registers in Broadcom's switching ASICs to accelerate memory diagnostics
- Developed an interprocess communication tool for use in Microsoft and OpenAI datacenters

Machine Learning Research Intern — September 2024 – June 2025
University of California, Santa Barbara — *Santa Barbara, California*

- Published a paper titled "Leveraging Large Language Models to Contextualize Network Measurements"
- Aggregated FCC data into a SQL database to improve model context using retrieval-augmented generation

Software Engineer Intern — July 2024 – September 2024
Cadence Design Systems / OpenEye Scientific — *Santa Fe, New Mexico*

- Wrote AWS-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated CUDA C++ software to incorporate new language features to match competing industry solutions

Hardware Architecture Intern — June 2023 – September 2023
Cadence Design Systems — *San Jose, California*

- Implemented a Python automation tool to generate memory wrappers for use on a hardware emulation platform
- Collected foundry provided memories written in System Verilog to be used as templates for this tool

Security Software Intern — June 2021 – August 2022
Axiado Corporation — *San Jose, California*

- Developed a Python package to create network packet flows for use in hardware simulation and emulation
- Created and executed a test plan for a hardware firewall's parser module and discovered multiple bugs

Technical Skills

Certificates: Enterprise Security Professional, Python for Data Science and AI
Languages: Python, C/C++, Java, JavaScript, Shell, Verilog
Developer Tools: Git, Visual Studio Code, Vim, Emacs
Libraries: Pandas, NumPy, PyTest, Black, Matplotlib